

General Linear Model

Info

Please select the aim of the analysis:

- **Calculate N** computes the required sample size given the **Target effect size** and **Minimum desired power**
- **Calculate Power** computes the achievable power given the **Target effect size** and **N (Sample size)**
- **Calculate Effect Size** computes the minimally-detectable effect size given the **N (Sample size)** and **Minimum desired power**

In all cases, you can set the required Type I error rate (significance cut-off)

Please notice that in case of multiple regression, the predictors are **assumed to be uncorrelated**. If correlated predictors are expected, please insert the correlations among covariates in the **Correlations panel** or use the **Partial Eta-squared** interface.

In all cases, you can set the required Type I error rate and whether the test will be carried out two-tailed or one-tailed.

For models with more than one independent variable, set the expected **R-squared** for the full model. For models with only one independent variable the R-squared is calculated as the square of the beta coefficients.

Set the **Model degrees of freedom**. If the model degrees of freedom are not easy to compute, please use the **Model definition** panel to help you out.

 Help can be found [Pamij web manual](#).

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Power parameters are computed for a general linear model with 1 degrees of freedom, for an effect size $\beta = 0.5$ with 1 degrees of freedom and type I error rate set to 0.05. The model is equivalent to a regression with 1 term or an ANOVA with 2 groups.

The required Sample size (N) is N=26.

A Priori Power Analysis

N	Effect size	f^2	Power	df	df(res)	α
26	0.500	0.333	0.800	1	24	0.0500

Note. When df=1 the R-square is the square of the beta coefficient.

Computed Parameters

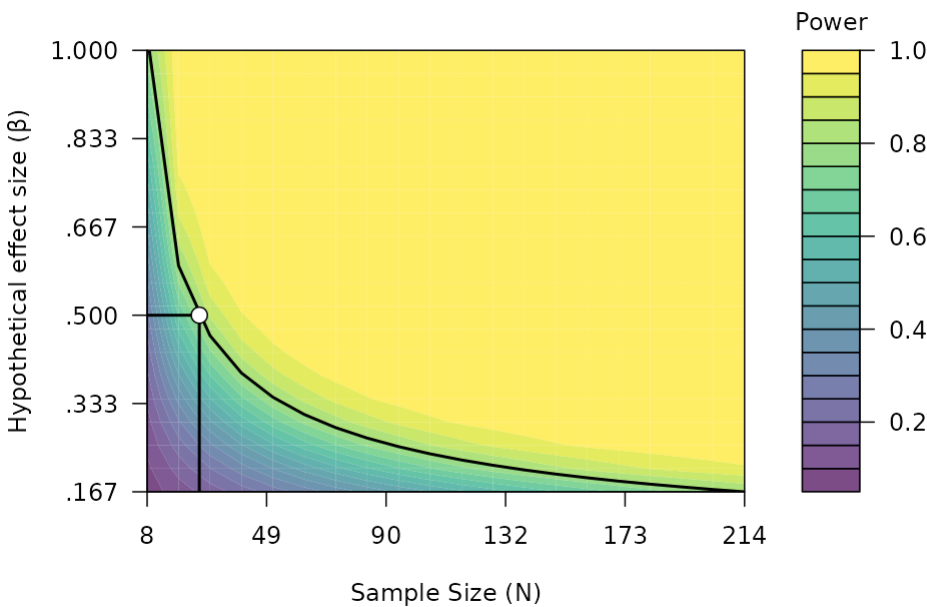
Index	Value
R ²	0.250
Effect tolerance	1.000
Effect η^2	0.250
Model df	1

Power by Effect Size

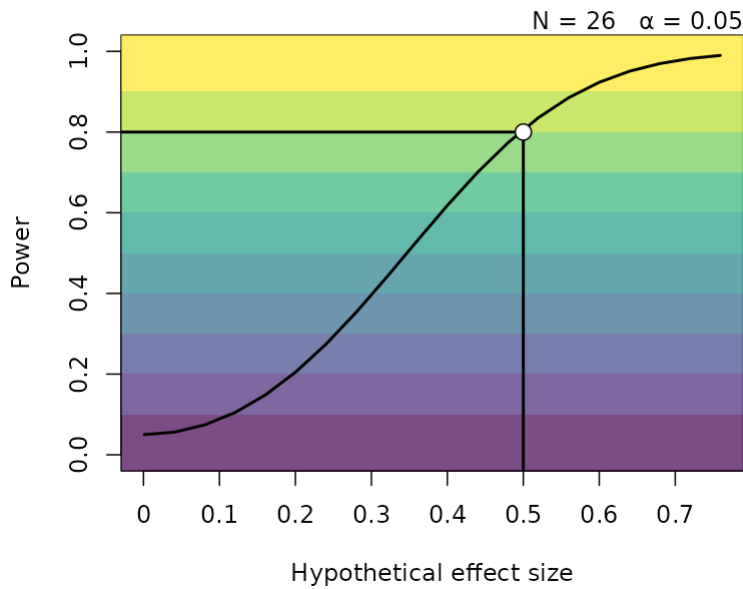
True effect size	Power to detect	Description
$0 < \beta \leq 0.347$	$\leq 50\%$	Likely miss
$0.347 < \beta \leq 0.496$	50% - 80%	Good chance of missing
$0.496 < \beta \leq 0.638$	80% - 95%	Probably detect
$\beta > 0.638$	$\geq 95\%$	Almost surely detect

Note. Estimated for N=26

Power Contour



Power Curve by Effect Size



Power Curve by N

